

Basic Disk Management Lab

Task 1 – Create MBR Partition with EXT4

Create a **1GB partition** on `/dev/sdb` using `fdisk`.

Requirements:

- Filesystem: **EXT4**
- Mount point: `/data`
- Must **automatically mount at boot**
- Use **UUID** in `/etc/fstab`

Verify:

```
df -h  
lsblk
```

Task 2 – Create GPT Partition with XFS

Initialize `/dev/sdc` using **GPT** and create a **2GB partition**.

Requirements:

Filesystem:

XFS

Mount point:

`/backup`

Add permanent mount entry in:

`/etc/fstab`

Verify:

```
mount -a  
df -h
```

Task 3 – Resize EXT4 Partition

The partition `/dev/sdb1` must be **expanded to 2GB**.

Requirements:

- Do not lose existing data
- Resize filesystem accordingly
- Mount should remain `/data`

Verify using:

```
lsblk  
df -h
```

Task 4 – Create LVM Storage

Create the following LVM structure using `/dev/sdd`.

Structure:

```
Physical Volume : /dev/sdd1  
Volume Group   : vgdata  
Logical Volume  : lvdata  
Size           : 2GB  
Filesystem      : EXT4  
Mount Point     : /lvdata
```

Add **permanent mount entry**.

Verify:

pvs
vgs
lvs

Task 5 – Create Multiple Logical Volumes

Inside **vgdata**, create two logical volumes.

Logical Volume Layout:

lvapps → 1GB → EXT4 → /apps
lvlogs → 1GB → XFS → /logs

All mounts must persist across reboot.

Verify:

lsblk
df -h

Task 6 – Extend Logical Volume (EXT4)

Extend **lvapps** by **1GB**.

Requirements:

- Filesystem must grow
- No unmount required if possible

Verify:

df -h /apps

Task 7 – Extend Logical Volume (XFS)

Increase **lvlogs** by **1GB**.

Requirements:

- Grow filesystem online

Verify:

`df -h /logs`

Task 8 – Create Volume Group with PE Size

Create a new **Volume Group with Physical Extent size of 8MB** using `/dev/sde`.

Structure:

VG name : vgbackup

PE size : 8MB

Verify using:

`vgdisplay`

Expected output should show:

PE Size = 8.00 MiB

Section 1 – Partitioning Labs (fdisk & parted)

1. Create an MBR Partition Using fdisk

- Initialize `/dev/sdb` with **MBR partition table**
- Create **1GB primary partition**
- Format with **EXT4**

- Mount at `/data1`
 - Add permanent mount entry in `/etc/fstab` using UUID
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2. Create Multiple MBR Partitions

- Using `fdisk`, create **three partitions** on `/dev/sdb`
- Sizes:
 - 500MB
 - 1GB
 - 1GB
- Format them with **EXT4**
- Mount at:

`/project`

`/backup`

`/archive`

- Configure **permanent mount**
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3. Create GPT Partition Using parted

- Initialize `/dev/sdc` with **GPT**
- Create **2GB partition**
- Format with **XFS**
- Mount at `/data2`

- Configure `/etc/fstab`
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4. Create Multiple GPT Partitions

Using `parted` on `/dev/sdc`:

Create

1GB

2GB

3GB

Filesystem types:

ext4

xfs

ext4

Mount points

/apps

/logs

/storage

Add permanent entries.

5. Delete and Recreate a Partition Without Losing Data

- Delete `/dev/sdb1`
 - Recreate it **using the same start sector**
 - Mount the filesystem and verify data integrity.
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6. Change Partition Type to LVM

- Create partition on `/dev/sdd`
- Change partition type to **LVM (8e)** using `fdisk`
- Verify using

lsblk
fdisk -l

7. Expand GPT Partition Using parted

- Increase `/dev/sdc1` from **2GB** to **4GB**
 - Grow filesystem.
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Section 2 – Filesystem Labs

8. Create EXT4 Filesystem with Custom Parameters

Create EXT4 filesystem on `/dev/sdb1` with:

block size = 4096
inode size = 256

Verify using

tune2fs -l

9. Create XFS Filesystem

- Create filesystem on `/dev/sdc1`
- Mount at `/xfldata`
- Verify using

xfs_info

10. Label Filesystems

Assign labels

DATA_EXT4

DATA_XFS

Mount using label instead of device.

11. Check Filesystem Integrity

Run filesystem checks on:

EXT4 → fsck

XFS → xfs_repair

Simulate corruption scenario (unmount required).



Section 3 – Disk Resize Labs

12. Increase EXT4 Partition Using fdisk

- Expand partition
- Resize filesystem using

resize2fs

Verify using

df -h

13. Decrease EXT4 Partition

- Shrink filesystem
 - Reduce partition size using fdisk
 - Mount again.
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14. Increase XFS Filesystem

- Expand partition
- Grow filesystem using

xfs_growfs

Note: **XFS cannot shrink**

15. Resize GPT Partition Using parted

- Resize `/dev/sdc1`
 - Extend filesystem online.
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Section 4 – LVM Basics

16. Create Complete LVM Setup

Create

PV → VG → LV

Structure

PV : `/dev/sdd1`

VG : `vgdata`

LV : `lvdata (2GB)`

Format with **EXT4**

Mount at `/lvdata`.

17. Create Two Logical Volumes

Inside `vgdata`

Create

lvext4 (2GB)

lvxfs (2GB)

Filesystem

ext4

xfs

Mount

`/lvext4`

`/lvxfs`

18. Create Volume Group with PE Size = 8MB

Create VG with

PE size = 8MB

Verify using

`vgdisplay`

19. Add Another Disk to Existing VG

Add `/dev/sde` to existing VG.

Verify using

`vgs`



Section 5 – LVM Resize Labs

20. Extend Logical Volume (EXT4)

Increase LV size by **1GB**

Resize filesystem.

21. Reduce Logical Volume (EXT4)

Shrink LV from **3GB** → **1GB**

Steps must include

umount
fsck
resize2fs
lvreduce
mount

22. Extend XFS Logical Volume

Increase LV by **2GB**

Grow filesystem online.

23. Extend Volume Group Using New Disk

Add **/dev/sdf** to VG.

Verify increased capacity.

Assume the system has these extra disks attached:

/dev/sdb
/dev/sdc
/dev/sdd
/dev/sde

You must complete all tasks using **root privileges**.

Section 1 – Basic Swap Tasks

Task 1 – Create Swap Partition Using fdisk

Create a **1GB swap partition** on **/dev/sdb**.

Requirements

- Use **fdisk**
- Change partition type to **Linux swap (82)**
- Enable swap immediately
- Verify swap usage
- Ensure swap **activates automatically after reboot**

Commands expected

```
fdisk  
mkswap  
swapon  
free -h
```

Add entry to:

/etc/fstab

Example format

```
UUID=<uuid> none swap defaults 0 0
```

Task 2 – Verify Active Swap

Check current swap configuration.

Commands

```
swapon --show  
free -h  
cat /proc/swaps
```

Section 2 – Swap File Tasks

Task 3 – Create Swap File (1GB)

Create swap file:

/swapfile

Requirements

- Size: **1GB**
- Correct permissions
- Activate swap
- Configure persistent entry

Commands expected

```
fallocate  
chmod  
mkswap  
swapon
```

Correct permissions:

```
chmod 600 /swapfile
```

fstab entry

```
/swapfile none swap defaults 0 0
```

Task 4 – Create Large Swap File (2GB)

Create swap file:

/swapdata

Requirements

- Size: **2GB**
- Activate immediately
- Verify swap usage.

Verification commands

```
swapon --show  
free -h
```

Section 3 – Swap Management Tasks

Task 5 – Disable Swap Temporarily

Disable all swap devices.

Commands

```
swapoff -a
```

Verify

```
free -h
```

Then enable swap again.

```
swapon -a
```

Task 6 – Remove Swap Partition

Remove swap from system.

Steps

1. Disable swap
2. Remove entry from `/etc/fstab`
3. Delete partition

Commands expected

`swapoff`

`fdisk`
